

Project Initialization and Planning Phase

Date	9 july 2026
Team ID	
Project Title	StudyAI-Smart AI educate Companion&Quiz Generator
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform how students prepare for exams by using AI to auto-generate study material, cutting revision time and improving retention. It tackles the inefficiency of manual note summarization, promising faster prep, better organization, and improved exam performance. Key features include an AI-powered summarization engine and an adaptive quiz/flashcard system.

Project Overview	
Objective	The primary objective is to help students study smarter by automatically converting uploaded study material into structured summaries, flashcards, and quizzes using AI.
Scope	The project covers the full student study workflow from file upload, through AI content generation, to quiz practice and performance tracking via an analytics dashboard.
Problem Statement	
Description	Students spend excessive time manually creating summaries, flashcards, and practice questions from their study material, leading to inefficient exam preparation.
Impact	Solving this will result in significant time savings, better-organized revision, improved exam readiness, and higher student confidence and performance.
Proposed Solution	
Approach	Using AI (Groq's Llama 3.3 70B via prompt engineering) to analyze uploaded documents and automatically generate structured summaries, flashcards, and quiz questions tailored to the content.
Key Features	- Implementation of an AI-based content generation engine for summaries, flashcards, and quizzes. - Real-time file processing for quick turnaround on uploaded notes.

	- Continuous analytics tracking to help students identify weak areas over time.
--	---

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Standard cloud CPU instance (no GPU needed- AI inference handled via Groq API, not local model)
Memory	RAM specifications	512 MB – 1 GB
Storage	Disk space for data, models, and logs	Minimal-no local model storage; only temporary file storage during upload
Software		
Frameworks	Python frameworks	Flask, Flask-JWT-Extended, Flask-CORS
Libraries	Additional libraries	Gunicorn, python-dotenv, requests
Development Environment	IDE	VS Code
Data		
Data	Source, size, format	User-uploaded study material no pre-existing dataset- content generated dynamically per upload via Groq API